SAP Business Data Cloud Connect

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SAP Business Data Cloud | SHIP

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Working with Data Products in SAP Business Data Cloud Connect

You can move from data to insights faster by securely enabling zero-copy, bidirectional data product sharing between SAP Business Data Cloud and supported external systems.

Make sure you have provisioned SAP Business Data Cloud Connect before sharing data products. See Provisioning SAP Business Data Cloud Connect.

Sharing Data Products from SAP Business Data Cloud Catalog through SAP Business Data Cloud Connect

You can make data products from SAP Business Data Cloud catalog available through SAP Business Data Cloud Connect. Sharing enables users to access and consume the data for analytics processing.

Sharing Data Products from SAP Partners to SAP Business Data Cloud

You can create a Delta Share to make your supported external system data available for downstream processing and collaboration. By publishing it as a data product with rich metadata, you can describe what your data means and how it's structured, so others can easily discover and use it. With this setup, you control the full lifecycle of your data products, from creating and updating shares, to publishing for consumption, and unpublishing when they're no longer needed.

SAP Business Data Cloud Connect Metering

SAP Business Data Cloud Connect metering provides information on cloud subscription usage for SAP public cloud products. Use it to understand and manage your subscriptions effectively, and to calculate the number of capacity units required.

Sharing Data Products from SAP Business Data Cloud Catalog through SAP Business Data Cloud Connect

You can make data products from SAP Business Data Cloud catalog available through SAP Business Data Cloud Connect. Sharing enables users to access and consume the data for analytics processing.

Prerequisites

To search for and share data products, you must have a global role that grants you the following privileges:

- **Catalog Asset** (–R——- -) - To access the catalog and view objects in the **Assets** and **Data Products** collections.
- **Cloud Data Product** (- - - - –S -) - To share data products to target systems.

You can apply the **Catalog Administrator** and the **BD Viewer** roles together, for example, to grant these privileges (see Standard Roles Delivered with SAP Business Data Cloud and Privileges and Permissions in SAP Business Data Cloud).

Context

In the central SAP Business Data Cloud catalog, you can select a data product to share to SAP BDC Connect.

Procedure

1. In the side navigation area, go to **Catalog & Marketplace** > **Search**.
2. On the catalog search page, use the filters or the search to find the data product you want and select it to open its details page. For more information, see Searching for Data Products and Assets in the Catalog.

   In the **Filter** panel, select the **Data Products** collection and the **Shareable** share status.

3. On the data product details page, choose the tab Overview > Details to review the list of APIs.

4. For the row you want, select the Share button to open the Manage Share Access dialog.

   If the data product has already been shared to other target systems, the Overview tab shows a list of those target systems. The Available Objects tab provides a list of the objects available.

   → **Tip**

   - If you're still viewing the search results in grid or table view, you can select Share for the data product.
   - If you're reviewing the API details page, you can select the Share button in the toolbar.

5. Choose Add Target.

6. Select the Databricks system that you want to share to.

7. Choose Share.

   As the share process runs, you'll receive a notification letting you know that the process started and a notification when it finishes with the result. You'll notice that the Manage Access dialog continually refreshes until the share process ends.

Results:

The data product is shared to the Databricks system, and the system is added to the list of target systems in the Overview tab.

Open Databricks to give users access to the data product. For details, review the Databricks help documentation . When users have access to a data product in Databricks, they can start consuming or enriching it.

If you need to change the access of the data product, you can change the access in the target system. No option is available for editing the share settings in the catalog. You can only delete the share.

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Sharing Data Products from SAP Partners to SAP Business Data Cloud

You can create a Delta Share to make your supported external system data available for downstream processing and collaboration. By publishing it as a data product with rich metadata, you can describe what your data means and how it's structured, so others can easily discover and use it. With this setup, you control the full lifecycle of your data products, from creating and updating shares, to publishing for consumption, and unpublishing when they're no longer needed.

Creating a Delta Share

Create a Delta Share in SAP Business Data Cloud Connect to publish data products with rich metadata. Use it to share and process data downstream.

Sharing Delta Share Data Products from SAP Partners to SAP Business Data Cloud

Sharing Delta Share data products with SAP partners enables secure, real-time collaboration through an open protocol that eliminates data duplication while ensuring high-performance access to a single source of truth.

Publishing the Delta Share as a Data Product to SAP Business Data Cloud

Set up your workspace and environment to create and manage shares. By installing the SDK and configuring shares, you describe your data in a way that makes it easy to share and reuse. Publishing the data product lets you make your data available for others to discover and use in SAP Business Data Cloud.

Unpublishing a Data Product

Unpublish a data product when you no longer want it to be accessed or used by others. This allows you to control when your data is available and ensure sharing stays aligned with your goals and policies. By unpublishing, you manage the full lifecycle of your data products and keep ownership over how they are used.

Appendix A: Describing the Data Product with ORD and CSN Metadata

With ORD, you provide details like title, description, tags, and links that help people discover and interpret your data product. With CSN, you define the schema—tables, columns, keys, and relationships—so users know exactly how to work with the data you are sharing.

Parent topic: Working with Data Products in SAP Business Data Cloud Connect

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Creating a Delta Share

Create a Delta Share in SAP Business Data Cloud Connect to publish data products with rich metadata. Use it to share and process data downstream.

Context

To expose a Delta Share for downstream processing in SAP BDC Connect, it needs to be published as a data product with rich metadata. This can be done using an iPython Notebook available in your SAP BDC Connect Workspace. For further details on how to describe the data product, see Appendix A: Describing the Data Product with ORD and CSN Metadata.

To create a Delta Share, follow the procedure as described in the partner's documentation:

Supported External Systems

Vendor	Hyperscaler	External Documentation
Databricks	AWS	Create a share object
	GCP	Create a share object
	Azure	Create a share object

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Sharing Delta Share Data Products from SAP Partners to SAP Business Data Cloud

Sharing Delta Share data products with SAP partners enables secure, real-time collaboration through an open protocol that eliminates data duplication while ensuring high-performance access to a single source of truth.

To share data assets with an SAP BDC recipient, follow the procedure as described in the partner's documentation:

Supported External Systems

Vendor	Hyperscaler	External Documentation
Databricks	AWS	Share Delta Sharing data shares with SAP BDC recipients
	GCP	Share Delta Sharing data shares with SAP BDC recipients
	Azure	Share Delta Sharing data shares with SAP BDC recipients

Parent topic: Sharing Data Products from SAP Partners to SAP Business Data Cloud

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Appendix A: Describing the Data Product with ORD and CSN Metadata

Publishing the Delta Share as a Data Product to SAP Business Data Cloud

Set up your workspace and environment to create and manage shares. By installing the SDK and configuring shares, you describe your data in a way that makes it easy to share and reuse. Publishing the data product lets you make your data available for others to discover and use in SAP Business Data Cloud.

Procedure

1. Install **sap-bdc-connect-sdk**:

    a. Run **pip install sap-bdc-connect-sdk** to install the python SDK.

    b. Verify the SDK installation by running **pip show sap-bdc-connect-sdk**. A successful output should look like this:

    Output Code

```
Name: sap-bdc-connect-sdk
Version: 1.0.9
Summary: Python SDK for SAP Business Data Cloud
Home-page: https://www.sap.com
Author: SAP SE
Author-email:
License: SAP DEVELOPER LICENSE AGREEMENT
Location: /local_disk0/.ephemeral_nfs/envs/pythonEnv-738b110e-411e-47f8-bf8e-355d6c2b
```

```
Requires: argparse, cryptography, grpcio-status, protobuf, pydantic, python-dateutil,
Required-by:
```

2. Create or update the share in **sap-bdc-connect-sdk**.

   A share is a mechanism for distributing and accessing data across different systems. Creating or updating a share involves including specific attributes, such as **@openResourceDiscoveryV1**, in the request body, aligning with the Open Resource Discovery protocol. This procedure ensures that the share is properly structured and described according to specified standards, facilitating effective data sharing and management.

   Copy the below code snippet and paste it to the cell, then provide the **share-name**, **title**, **shortDescription** and **description** of your share.

   ### ℹ Note

   Ensure that the **shortDescription** and **description** are not the same.

   ### Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

open_resource_discovery_information = {
    "@openResourceDiscoveryV1": {
        "title": "<title>",
        "shortDescription": "<short-description>",
        "description": "<description>"
    }
}

bdc_connect_client.create_or_update_share(
    share_name,
    open_resource_discovery_information
)
```

   ### → Tip

   To get the <recipient-name>, go to **Databricks Workspace** > **Catalog** > **Manage (Gear icon)** > **Delta Sharing** > **Shared by me** > **Recipients** >. The recipient name is under the field **Name**.

3. Create or update share CSN.

   The CSN serves as a standardized format for configuring and describing shares within a network. To create or update the CSN for a share, it's advised to prepare the CSN content in a separate file and include this content in the request body. This approach ensures accuracy and compliance with the CSN interoperability specifications, facilitating consistent and effective share configuration across systems.

   Copy the below code snippet and paste it to the cell, then provide the share-name of your share.

   ### ℹ Note

   The CSN schema can be efficiently generated by using the function `generate_csn_template()` available the in `csn_generator` file. This script automates the creation of CSN content for Databricks environments based on a Databricks share.

   ### Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient
from bdc_connect_sdk.utils import csn_generator

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

csn_schema = csn_generator.generate_csn_template(share_name)

bdc_connect_client.create_or_update_share_csn(
    share_name,
    csn_schema
)
```

→ Tip

To get the <recipient-name>, go to Databricks Workspace > Catalog > Manage (Gear icon) > Delta Sharing > Shared by me > Recipients. The recipient name is under the field Name.

If the generated CSN needs to be modified for any reason, it can be written to a file for editing, for example:

Sample Code

```
from bdc_connect_sdk.utils import csn_generator
import json

share_name = "<share-name>"
csn_schema = csn_generator.generate_csn_template(share_name)

file_path = f"csn_{share_name}.json"
with open(file_path, "w") as csn_file:
    csn_file.write(json.dumps(csn_schema, indent=2))

# change the file

with open(file_path, "r") as csn_file:
    changed_csn_schema = csn_file.read()

csn_schema = changed_csn_schema
```

4. Publish a data product.

   A data product is an abstraction that represents a type of data or data set within a system, facilitating the management and sharing of data products across different platforms. It bundles resources or API endpoints to enable efficient data access and utilization by integrated systems. Publishing a data product allows these systems to access and consume the data, ensuring seamless communication and resource sharing.

   Copy the below code snippet and paste it to the cell, then provide the share-name of your share. This will publish the share as a data product for consumption by other services in SAP Business Data Cloud.

Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

bdc_connect_client.publish_data_product(
    share_name
)
```

→ Tip

To get the <recipient-name>, go to **Databricks Workspace** > **Catalog** > **Manage (Gear icon)** > **Delta Sharing** > **Shared by me** > **Recipients** >. The recipient name is under the field **Name**.

i Note

The published data product will not appear instantly in SAP Datasphere, since it starts a list of actions that take place in pre-scheduled intervals. If you don't see your data products, wait for a few minutes and try again.

Once a data product has been published from SAP BDC Connect to SAP Business Data Cloud, you can use the catalog **Data Product** collection to view data products for use in your modeling and other projects. For more information, see Evaluating and Installing Data Products.

Task overview: Sharing Data Products from SAP Partners to SAP Business Data Cloud

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Unpublishing a Data Product

Unpublish a data product when you no longer want it to be accessed or used by others. This allows you to control when your data is available and ensure sharing stays aligned with your goals and policies. By unpublishing, you manage the full lifecycle of your data products and keep ownership over how they are used.

Context

Unpublishing a data product involves withdrawing access to the data, typically when it has been sufficiently consumed or is no longer desired to be shared. This action helps maintain control over data accessibility and ensures that sharing aligns with security and usage policies. By unpublishing, the system can effectively manage the lifecycle of data and its availability to other integrated platforms.

Procedure

1. Copy the below code snippet and paste it to the cell:

Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

bdc_connect_client.delete_share(
    share_name
)
```

→ Tip

To get the <recipient-name>, go to **Databricks Workspace** > **Catalog** > **Manage (Gear icon)** > **Delta Sharing** > **Shared by me** > **Recipients** >. The recipient name is under the field **Name**.

2. Provide the **share-name** of your share.
3. Run the script.

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Appendix A: Describing the Data Product with ORD and CSN Metadata

With ORD, you provide details like title, description, tags, and links that help people discover and interpret your data product. With CSN, you define the schema—tables, columns, keys, and relationships—so users know exactly how to work with the data you are sharing.

A data product must be described in ORD (Open Resource Discovery) and CSN (Core Schema Notation), compliant with CSN Interop Specification v1.0 https://sap.github.io/csn-interop-specification/ .

The ORD information allows you to describe the content of the data product using text, links, tags, labels, etc. The CSN definition intends to describe the schema of the data, specifying the tables being shared, and their columns with the definition of data types, primary keys, associations/references, etc.

ORD Data Product Definition

The ORD data product schema contains the complete list of attributes that compose the ORD data product metadata. However, not all fields can be set by users.

- You must provide these fields when publishing data products, as they are mandatory in the data product object, and the provider system cannot generate them automatically.

Field	Mandatory	Example Value
title	yes	"Customer Order"
shortDescription	yes	"Offering access to all online and offline orders submitted by customers."
description	yes	"The data product Customer Order offers access to all online and offline orders submitted by customers. It provides a customer view of the orders. For fulfillment-specific aspects please refer to the data product Fulfillment Order."

- You must define these optional fields in the data product object if you want them included, as the provider system will not generate them automatically and they will not be present by default in the resulting data product.

Field	Mandatory	Example Value
labels	false	Sample Code {
deprecationDate	false	Sample Code "2025-12-19T15:47:04+00:00"
sunsetDate	false	Sample Code "2026-12-19T15:47:04+00:00"
successors	false	Sample Code ["customer.ext:dataProduct:other:v1"]
industry	false	Sample Code ["Retail", "Public Sector"]
lineOfBusiness	false	Sample Code ["Sales"]
tags	false	Sample Code ["storage", "high-availability"]
correlationIds	false	Sample Code ["sap.s4:communicationScenario:SAP_COM_0008"]
dataProductLinks	false	Sample Code [

Field	Mandatory	Example Value
links	false	Sample Code [
documentationLabels	false	Sample Code {

- You can overwrite the following fields:

Field	Mandatory	Example Value	Comments
visibility	yes	"public"	Default value: "public". Alternative values: "interval", "private."
releaseStatus	yes	"active"	Default value: "active". Alternative values: "beta", "deprecated."

The minimum required fields that need to be specified are title , shortDescription , and description .

Note

The **description** value must not contain the **shortDescription** value.

You can describe the ORD attributes for your data product on a JSON document. This is an example of how the ORD information JSON document should look like:

Sample Code

```
{
"title": "TPC-H",
"shortDescription": "TPC-H is a decision support
"description": "TPC-H consists of a suite of bus
}
```

CSN Interop Definition

In addition to the ORD metadata, the CSN Interop metadata is also required when publishing a data product. For a complete documentation of the CSN Interop specification, see CSN Interop Effective v1 • Interface Documentation . For examples of CSN documents compliant with this specification, see CSN Interop Effective v1 • Examples .

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SAP Business Data Cloud Connect Metering

SAP Business Data Cloud Connect metering provides information on cloud subscription usage for SAP public cloud products. Use it to understand and manage your subscriptions effectively, and to calculate the number of capacity units required.

To calculate the number of capacity units required for each SAP BDC capacity service, see SAP Business Data Cloud Capacity Unit Estimator.

To learn about your subscriptions across your SAP public cloud products in the **Public Cloud Consumption** card, see Public Cloud Consumption Card.

For information on cloud subscription usage, go to the page SAP Cloud Subscription Usage and search for **Capacity Units - SAP Business Data Cloud** (CY475). Choose the link **SAP Business Data Cloud**. In the PDF document that opens, search for **Capacity Units – SAP Business Data Cloud Connect** (CY555).

For SAP BDC Connect capacity services, go to the page Cloud Services. Use the search bar to find the card **SAP Business Data Cloud Supplement**. Choose a card in one of the available languages. In that document, under the section **ATTACHMENT A**, search for **Table 5: SAP Business Data Cloud Connect Capacity Services**.

Parent topic: Working with Data Products in SAP Business Data Cloud Connect

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Requesting Help from SAP Technical Support

You can count on SAP Technical Support to help you resolve SAP Business Data Cloud issues when they arise. Following this procedure makes it easier for support engineers to understand your case and provide the right solution faster.

Prerequisites

Before creating a case, make sure you have:

- Checked the availability by service and region in the platform status. See SAP Trust Center.
- Reviewed the guidelines on how to engage with support. See Customer Support Best Practices and Offerings.

- Checked what to provide in an SAP BDC case and created a support user with the appropriate roles to allow an SAP support engineer to log in and troubleshoot your system. See SAP Note 3568017.

Procedure

To report a case to SAP Technical Support using SAP for Me, follow these steps:

1. Open SAP for Me and log in.
2. Open the **Services & Support** dashboard.
3. (Optional) You can use the **Knowledge Search** tab to find existing solutions in the following resources:
   - SAP Notes & KBAs
   - SAP Community
   - SAP Support Portal
   - Product Documentation
4. Use the **Get Support (Case, Expert Chat...)** card within SAP for Me to get assistance via the available support channels.

   For more information on how to report technical issues, see Get Support Application.

   You can receive SAP BDC service status notifications. See Manage Cloud Service Availability Notifications.

   For non-technical assistance, see Customer Interaction Center.

Related Information

SAP Business Data Cloud Community
Latest News on SAP Business Data Cloud